

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–18

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

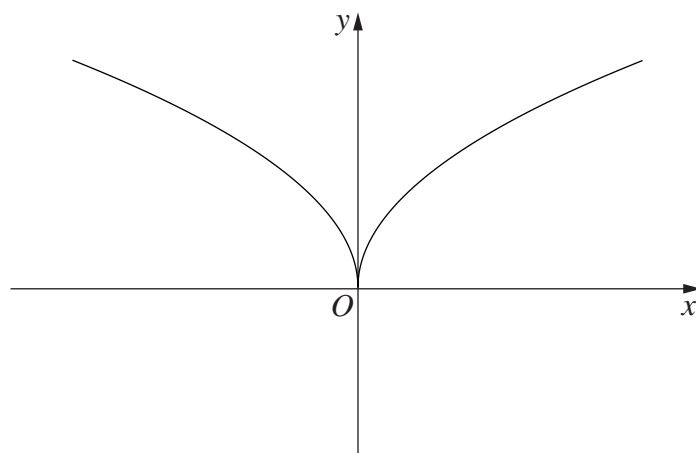
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Which equation best represents the following graph?

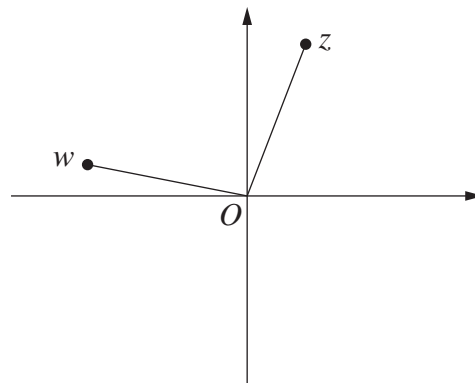


- (A) $y = \sqrt{x}$
(B) $|y| = \sqrt{x}$
(C) $y = \sqrt{|x|}$
(D) $|y| = \sqrt{|x|}$
- 2 Which polynomial has a multiple root at $x = 1$?
- (A) $x^5 - x^4 - x^2 + 1$
(B) $x^5 - x^4 - x - 1$
(C) $x^5 - x^3 - x^2 + 1$
(D) $x^5 - x^3 - x + 1$

- 3 The sum of the eccentricities of two different conics is $\frac{3}{4}$.

Which pair of conics could this be?

- (A) Circle and ellipse
 - (B) Ellipse and parabola
 - (C) Parabola and hyperbola
 - (D) Hyperbola and circle
- 4 The Argand diagram shows the complex numbers z and w , where z lies in the first quadrant and w lies in the second quadrant.



Which complex number could lie in the 3rd quadrant?

- (A) $-w$
- (B) $2iz$
- (C) $\bar{z}$
- (D) $w - z$

- 5** Multiplying a non-zero complex number by $\frac{1-i}{1+i}$ results in a rotation about the origin on an Argand diagram.

What is the rotation?

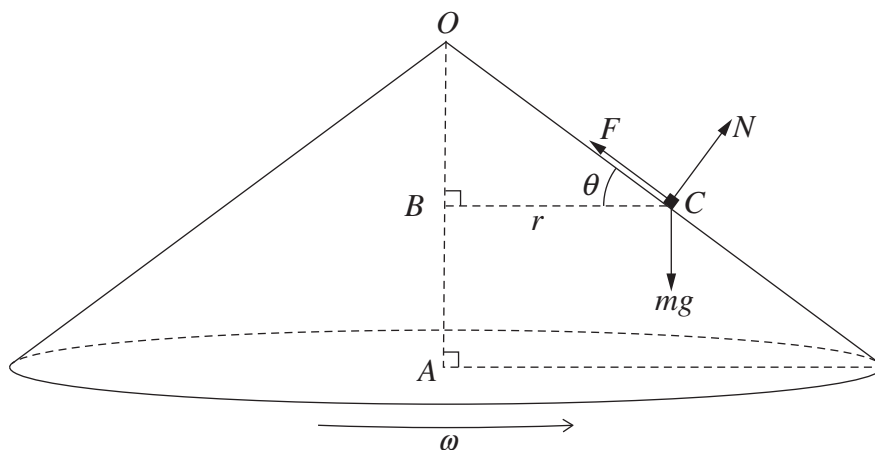
- (A) Clockwise by $\frac{\pi}{4}$
- (B) Clockwise by $\frac{\pi}{2}$
- (C) Anticlockwise by $\frac{\pi}{4}$
- (D) Anticlockwise by $\frac{\pi}{2}$
- 6** Let $p(x) = 1 + x + x^2 + x^3 + \cdots + x^{12}$.
- What is the coefficient of x^8 in the expansion of $p(x+1)$?
- (A) 1
- (B) 495
- (C) 715
- (D) 1287
- 7** The hyperbola with equation $xy = 8$ is the hyperbola $x^2 - y^2 = k$ referred to different axes.

What is the value of k ?

- (A) 2
- (B) 4
- (C) 8
- (D) 16

- 8 A small object of mass m kg sits on a rotating conical surface at C , r metres from the axis OA and with $\angle OCB = \theta$, as shown in the diagram.

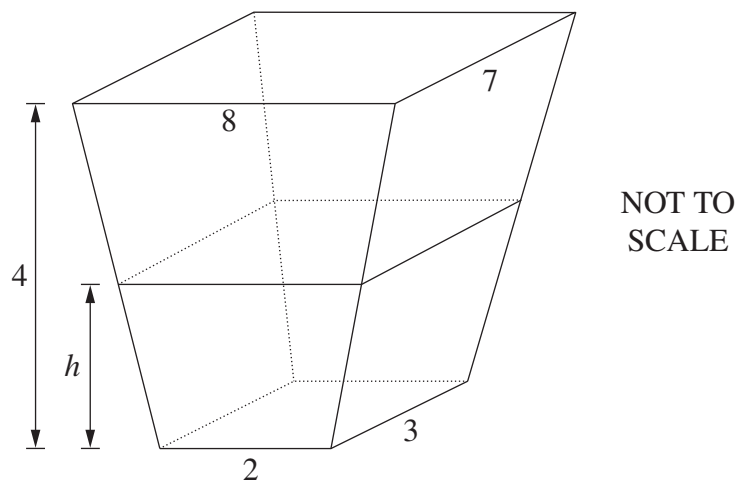
The surface is rotating about its axis with angular velocity ω . The forces acting on the object are gravity, a normal reaction force N and a frictional force F , which prevents the object from sliding down the surface.



Which pair of statements is correct?

- (A) $F \cos \theta + N \sin \theta = mr\omega^2$
 $F \sin \theta + N \cos \theta = mg$
- (B) $F \cos \theta + N \sin \theta = mr\omega^2$
 $F \sin \theta - N \cos \theta = mg$
- (C) $F \cos \theta - N \sin \theta = mr\omega^2$
 $F \sin \theta + N \cos \theta = mg$
- (D) $F \cos \theta - N \sin \theta = mr\omega^2$
 $F \sin \theta - N \cos \theta = mg$

- 9 The diagram shows the dimensions of a polyhedron with parallel base and top. A slice taken at height h parallel to the base is a rectangle.



What is a correct expression for the volume of the polyhedron?

- (A) $\int_0^4 (h+3) \left(\frac{3h}{2} + 2 \right) dh$
- (B) $\int_0^4 \left(\frac{5h}{4} + 3 \right) \left(\frac{3h}{2} + 2 \right) dh$
- (C) $\int_0^4 (h+3) \left(\frac{5h}{4} + 2 \right) dh$
- (D) $\int_0^4 \left(\frac{5h}{4} + 3 \right) \left(\frac{5h}{4} + 2 \right) dh$
- 10 Suppose that $x + \frac{1}{x} = -1$.

What is the value of $x^{2016} + \frac{1}{x^{2016}}$?

- (A) 1
- (B) 2
- (C) $\frac{2\pi}{3}$
- (D) $\frac{4\pi}{3}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

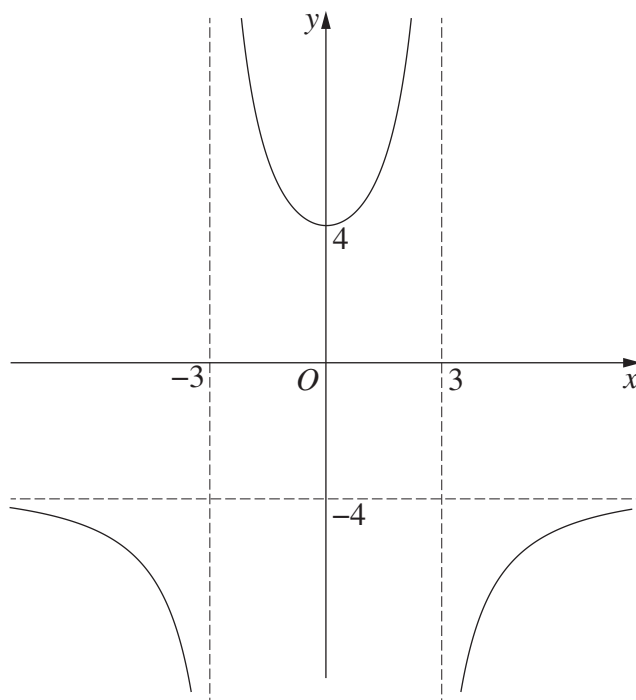
Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Let $z = \sqrt{3} - i$.
- | | |
|--|----------|
| (i) Express z in modulus–argument form. | 2 |
| (ii) Show that z^6 is real. | 1 |
| (iii) Find a positive integer n such that z^n is purely imaginary. | 1 |
-
- (b) Find $\int x e^{-2x} dx$.
- 3**
-
- (c) Find $\frac{dy}{dx}$ for the curve given by $x^3 + y^3 = 2xy$, leaving your answer in terms of x and y .
- 2**

Question 11 continues on page 8

Question 11 (continued)

- (d) The diagram shows the graph of $y = f(x)$.



Draw a separate half-page diagram for each of the following functions, showing all asymptotes and intercepts.

(i) $y = \sqrt{f(x)}$ 2

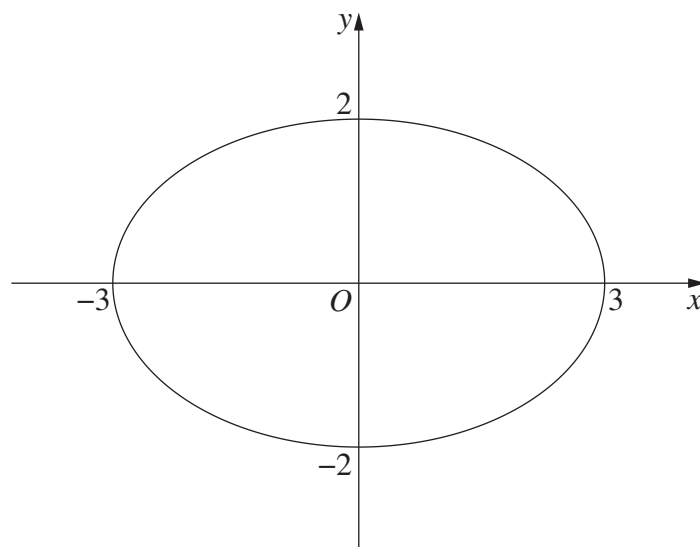
(ii) $y = \frac{1}{f(x)}$ 2

- (e) State the domain and range of the function $f(x) = x \sin^{-1}\left(\frac{x}{2}\right)$. 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) The diagram shows an ellipse.



- | | | |
|-------|--|---|
| (i) | Write an equation for the ellipse. | 1 |
| (ii) | Find the eccentricity of the ellipse. | 1 |
| (iii) | Write the coordinates of the foci of the ellipse. | 1 |
| (iv) | Write the equations of the directrices of the ellipse. | 1 |
-
- | | | | |
|-----|------|---|---|
| (b) | (i) | Differentiate $xf(x) - \int x f'(x) dx$. | 1 |
| | (ii) | Hence, or otherwise, find $\int \tan^{-1} x dx$. | 2 |

Question 12 continues on page 10

Question 12 (continued)

(c) Let $z = \cos\theta + i\sin\theta$.

(i) By considering the real part of z^4 , show that $\cos 4\theta$ is **2**

$$\cos^4\theta - 6\cos^2\theta \sin^2\theta + \sin^4\theta.$$

(ii) Hence, or otherwise, find an expression for $\cos 4\theta$ involving only powers of $\cos\theta$. **1**

(d) (i) Show that the equation of the normal to the hyperbola $xy = c^2$, $c \neq 0$, at $P\left(cp, \frac{c}{p}\right)$ is given by $px - \frac{y}{p} = c\left(p^2 - \frac{1}{p^2}\right)$. **2**

(ii) The normal at P meets the hyperbola again at $Q\left(cq, \frac{c}{q}\right)$. **3**

Show that $q = -\frac{1}{p^3}$.

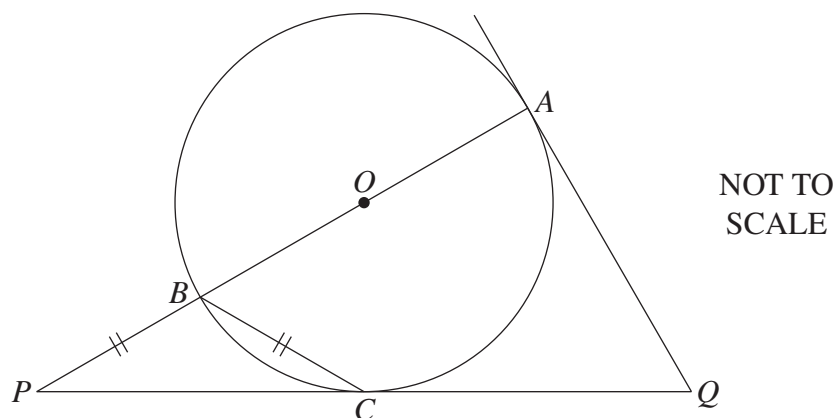
End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The function $f(x) = x^x$ is defined and positive for all $x > 0$. **3**

By differentiating $\ln(f(x))$, find the value of x at which $f(x)$ has a minimum.

- (b) The circle centred at O has diameter AB . A point P on AB produced is chosen so that PC is a tangent to the circle at C and $BP = BC$. The tangents to the circle at A and C meet at Q . **4**



Copy or trace the diagram into your writing booklet.

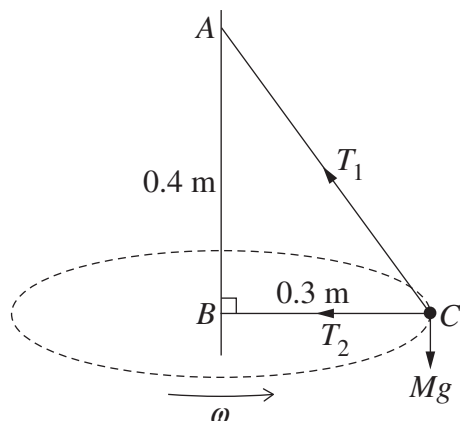
Prove that $OP = OQ$.

Question 13 continues on page 12

Question 13 (continued)

- (c) The ends of a string are attached to points A and B , with A directly above B . The points A and B are 0.4 m apart.

An object of mass M kg is fixed to the string at C . The object moves in a horizontal circle with centre B and radius 0.3 m, as shown in the diagram.



The tensions in the string from the object to points A and B are T_1 and T_2 respectively. The object rotates with constant angular velocity ω . You may assume that the acceleration due to gravity is $g = 10 \text{ m s}^{-2}$.

- (i) Show that $T_2 = 0.3M(\omega^2 - 25)$. 3
- (ii) For what range of values of ω is $T_2 > T_1$? 1
- (d) Suppose $p(x) = ax^3 + bx^2 + cx + d$ with a, b, c and d real, $a \neq 0$.
- (i) Deduce that if $b^2 - 3ac < 0$ then $p(x)$ cuts the x -axis only once. 2
- (ii) If $b^2 - 3ac = 0$ and $p\left(-\frac{b}{3a}\right) = 0$, what is the multiplicity of the root $x = -\frac{b}{3a}$? 2

End of Question 13

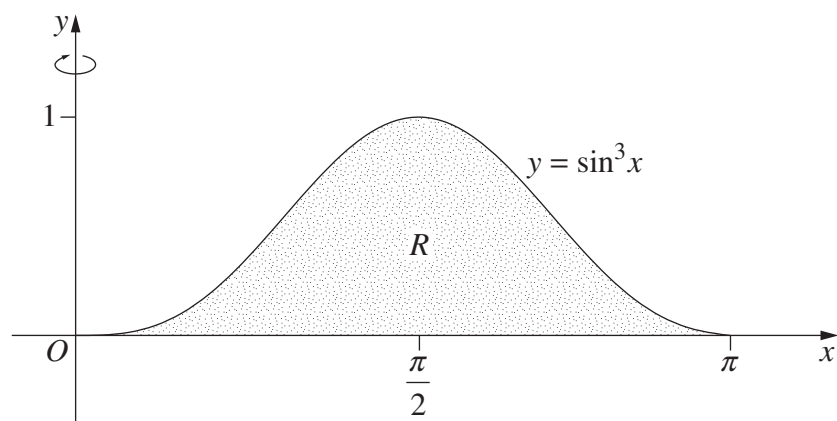
Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Show that $\int \sin^3 x \, dx = \frac{1}{3} \cos^3 x - \cos x + C$. **1**

(ii) Using a graphical approach, or otherwise, explain why **1**

$$\int_0^\pi \cos^{2n-1} x \, dx = 0, \text{ for all positive integers } n.$$

(iii) The diagram shows the region R enclosed by $y = \sin^3 x$ and the x -axis for $0 \leq x \leq \pi$. **3**



Using the method of cylindrical shells and the results in parts (i) and (ii), find the exact volume of the solid formed when R is rotated about the y -axis.

Question 14 continues on page 14

Question 14 (continued)

(b) Let $I_n = \int_0^1 \frac{x^n}{(x^2 + 1)^2} dx$, for $n = 0, 1, 2, \dots$.

(i) Using a suitable substitution, show that $I_0 = \frac{\pi}{8} + \frac{1}{4}$. **3**

(ii) Show that $I_0 + I_2 = \frac{\pi}{4}$. **1**

(iii) Find I_4 . **3**

(c) Show that $x\sqrt{x} + 1 \geq x + \sqrt{x}$, for $x \geq 0$. **3**

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) The equation $x^3 - 3x + 1 = 0$ has roots α, β and γ . **2**

Find a cubic equation with integer coefficients that has roots α^2, β^2 and γ^2 .

- (b) A particle is initially at rest at the point B which is b metres to the right of O . The particle then moves in a straight line towards O .

For $x \neq 0$, the acceleration of the particle is given by $\frac{-\mu^2}{x^2}$, where x is the distance from O and μ is a positive constant.

- (i) Prove that $\frac{dx}{dt} = -\mu\sqrt{2}\sqrt{\frac{b-x}{bx}}$. **2**

- (ii) Using the substitution $x = b\cos^2\theta$, show that the time taken to reach a distance d metres to the right of O is given by **3**

$$t = \frac{b\sqrt{2b}}{\mu} \int_0^{\cos^{-1}\sqrt{\frac{d}{b}}} \cos^2\theta d\theta.$$

It can be shown that $t = \frac{1}{\mu}\sqrt{\frac{b}{2}}\left(\sqrt{bd-d^2} + b\cos^{-1}\sqrt{\frac{d}{b}}\right)$. (Do NOT prove this.)

- (iii) What is the limiting time taken for the particle to reach O ? **1**

Question 15 continues on page 16

Question 15 (continued)

- (c) (i) Use partial fractions to show that **2**

$$\frac{3!}{x(x+1)(x+2)(x+3)} = \frac{1}{x} - \frac{3}{x+1} + \frac{3}{x+2} - \frac{1}{x+3}.$$

- (ii) Suppose that for n a positive integer **3**

$$\frac{n!}{x(x+1)\dots(x+n)} = \frac{a_0}{x} + \frac{a_1}{x+1} + \dots + \frac{a_k}{x+k} + \dots + \frac{a_n}{x+n}.$$

Show that $a_k = (-1)^k \binom{n}{k}$.

- (iii) Hence, or otherwise, find the limiting sum of **2**

$$1 - \frac{1}{2} \binom{n}{1} + \frac{1}{3} \binom{n}{2} - \frac{1}{4} \binom{n}{3} + \dots + \frac{(-1)^n}{n+1}.$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) The complex numbers $z = \cos\theta + i\sin\theta$ and $w = \cos\alpha + i\sin\alpha$, where $-\pi < \theta \leq \pi$ and $-\pi < \alpha \leq \pi$, satisfy **3**

$$1 + z + w = 0.$$

By considering the real and imaginary parts of $1 + z + w$, or otherwise, show that $1, z$ and w form the vertices of an equilateral triangle in the Argand diagram.

- (ii) Hence, or otherwise, show that if the three non-zero complex numbers $2i, z_1$ and z_2 satisfy **2**

$$|2i| = |z_1| = |z_2| \quad \text{AND} \quad 2i + z_1 + z_2 = 0$$

then they form the vertices of an equilateral triangle in the Argand diagram.

- (b) (i) The complex numbers $0, u$ and v form the vertices of an equilateral triangle in the Argand diagram. **2**

Show that $u^2 + v^2 = uv$.

- (ii) Give an example of non-zero complex numbers u and v , so that $0, u$ and v form the vertices of an equilateral triangle in the Argand diagram. **1**

Question 16 continues on page 18

Question 16 (continued)

- (c) In a group of n people, each has one hat, giving a total of n different hats. They place their hats on a table. Later, each person picks up a hat, not necessarily their own.

A situation in which none of the n people picks up their own hat is called a derangement.

Let $D(n)$ be the number of possible derangements.

- (i) Tom is one of the n people. In some derangements Tom finds that he and one other person have each other's hat. 1

Show that, for $n > 2$, the number of such derangements is $(n-1)D(n-2)$.

- (ii) By also considering the remaining possible derangements, show that, for $n > 2$, 2

$$D(n) = (n-1)[D(n-1) + D(n-2)].$$

- (iii) Hence, show that $D(n) - nD(n-1) = -[D(n-1) - (n-1)D(n-2)]$, for $n > 2$. 1

- (iv) Given $D(1) = 0$ and $D(2) = 1$, deduce that $D(n) - nD(n-1) = (-1)^n$, for $n > 1$. 1

- (v) Prove by mathematical induction, or otherwise, that for all integers $n \geq 1$, 2

$$D(n) = n! \sum_{r=0}^n \frac{(-1)^r}{r!}.$$

End of paper

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REFERENCE SHEET

– Mathematics –

– Mathematics Extension 1 –

– Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

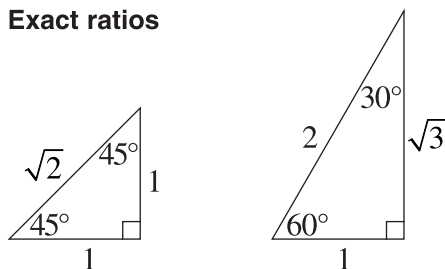
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

tangent: $y = tx - at^2$

normal: $x + ty = at^3 + 2at$

At (x_1, y_1) ,

tangent: $xx_1 = 2a(y + y_1)$

normal: $y - y_1 = -\frac{2a}{x_1}(x - x_1)$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

2016 HSC Mathematics Extension 2 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	C
2	C
3	A
4	D
5	B
6	C
7	D
8	C
9	A
10	B

Section II

Question 11 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Provides exact argument in radians, or equivalent merit	1

Sample answer:

$$z = \sqrt{3} - i$$

$$|z| = 2, \quad \text{Arg}(z) = -\frac{\pi}{6}$$

$$\text{So } z = 2 \left(\cos\left(\frac{-\pi}{6}\right) + i \sin\left(\frac{-\pi}{6}\right) \right)$$

Question 11 (a) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\begin{aligned} z^6 &= 2^6 (\cos(-\pi) + i \sin(-\pi)) \\ &= -2^6 \quad \text{which is real} \end{aligned}$$

Question 11 (a) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

Take $n = 3$

$$\begin{aligned} z^3 &= 2^3 \left(\cos\left(\frac{-\pi}{2}\right) + i \sin\left(\frac{-\pi}{2}\right) \right) \\ &= -2^3 i \quad \text{which is purely imaginary} \end{aligned}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	3
• Correctly applies integration by parts, or equivalent merit	2
• Attempts to use integration by parts, or equivalent merit	1

Sample answer:

$$\int x e^{-2x} dx \quad u = x, \quad \frac{dv}{dx} = e^{-2x}$$

$$= \frac{x e^{-2x}}{-2} + \frac{1}{2} \int e^{-2x} dx$$

$$= -\frac{x e^{-2x}}{2} - \frac{1}{4} e^{-2x} + C$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Correctly differentiates xy , or equivalent merit	1

Sample answer:

$$x^3 + y^3 = 2xy$$

Differentiating with respect to x ,

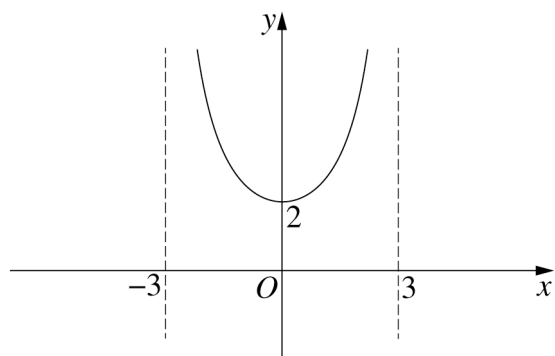
$$3x^2 + 3y^2 y' = 2xy' + 2y$$

$$y'(3y^2 - 2x) = 2y - 3x^2$$

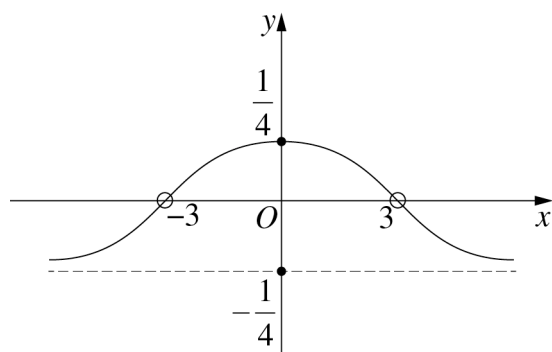
$$y' = \frac{2y - 3x^2}{3y^2 - 2x}$$

Question 11 (d) (i)

Criteria	Marks
• Provides correct graph	2
• Recognises correct domain, or equivalent merit	1

Sample answer:**Question 11 (d) (ii)**

Criteria	Marks
• Provides correct graph	2
• Correctly deals with the asymptotes, or equivalent merit	1

Sample answer:

Question 11 (e)

Criteria	Marks
• Provides correct solution	2
• Provides correct domain, or equivalent merit	1

Sample answer:

$$f(x) = x \sin^{-1}\left(\frac{x}{2}\right)$$

Domain: $-2 \leq x \leq 2$ Range: $0 \leq f(x) \leq \pi$ **Question 12 (a) (i)**

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$

Question 12 (a) (ii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{4}{9}} = \frac{\sqrt{5}}{3}$$

Question 12 (a) (iii)

Criteria	Marks
• Provides correct answer	1

*Sample answer:*Foci at $(\pm ae, 0) = (\pm\sqrt{5}, 0)$

Question 12 (a) (iv)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\text{Directrices are } x = \pm \frac{a}{e} = \pm \frac{9}{\sqrt{5}}$$

Question 12 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$y = x f(x) - \int x f'(x) dx$$

$$\begin{aligned} \frac{dy}{dx} &= f(x) + x f'(x) - x f'(x) \\ &= f(x) \end{aligned}$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to apply result from part (i), or equivalent merit	1

Sample answer:

$$\begin{aligned} \int \tan^{-1} x dx &= x \tan^{-1} x - \int x \frac{1}{1+x^2} dx \quad (\text{From part (i)}) \\ &= x \tan^{-1} x - \int \frac{x}{1+x^2} dx \\ &= x \tan^{-1} x - \frac{1}{2} \log(1+x^2) + k \end{aligned}$$

Question 12 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Applies de Moivre's theorem, or equivalent merit	1

Sample answer:

$$z = \cos \theta + i \sin \theta$$

$$z^4 = \cos 4\theta + i \sin 4\theta \quad (\text{de Moivre's theorem})$$

$$z^4 = (\cos \theta + i \sin \theta)^4$$

 $\therefore$ equating real parts,

$$\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

Question 12 (c) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$\text{using } \sin^2 \theta = 1 - \cos^2 \theta$$

$$\begin{aligned} \therefore \cos 4\theta &= \cos^4 \theta - 6\cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2 \\ &= 8\cos^4 \theta - 8\cos^2 \theta + 1 \end{aligned}$$

Alternative solution

$$\begin{aligned} \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ &= 2\cos^2 \theta - 1 \end{aligned}$$

$$\begin{aligned} \therefore \cos 4\theta &= 2\cos^2 2\theta - 1 \\ &= 2(2\cos^2 \theta - 1)^2 - 1 \\ &= 8\cos^4 \theta - 8\cos^2 \theta + 1 \end{aligned}$$

Question 12 (d) (i)

Criteria	Marks
• Provides correct solution	2
• Finds the slope of the normal at P , or equivalent merit	1

Sample answer:

$$xy = c^2$$

$$\text{d.w.r.t. } x \quad y + x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-y}{x}$$

$$= \frac{-c}{cp^2} = \frac{-1}{p^2}$$

Slope of normal at P is p^2 .

So equation of normal is $y = p^2x + k$.

Since the normal passes through P

$$\frac{c}{p} = p^2 \times cp + k$$

$$k = \frac{c}{p} - cp^3$$

$$\therefore y = p^2x + \frac{c}{p} - cp^3$$

$$\frac{y}{p} = px + \frac{c}{p^2} - cp^2 \quad (\div p)$$

$$\therefore px - \frac{y}{p} = c \left(p^2 - \frac{1}{p^2} \right)$$

Question 12 (d) (ii)

Criteria	Marks
• Provides correct solution	3
• Obtains a quadratic whose roots are cp and cq , or equivalent merit	2
• Attempts to solve the equations for the normal and hyperbola, or equivalent merit	1

Sample answer:

Solving $xy = c^2$ and the equation of the normal,

$$px - \frac{c^2}{px} = c \left(p^2 - \frac{1}{p^2} \right)$$

$$\therefore p^2 x^2 - pcx \left(p^2 - \frac{1}{p^2} \right) - c^2 = 0$$

The roots are cp and cq so using their product,

$$c^2 pq = -\frac{c^2}{p^2}$$

$$\therefore q = -\frac{1}{p^3}$$

Question 13 (a)

Criteria	Marks
• Provides correct solution	3
• Finds the value of x at which there is a stationary point, or equivalent merit	2
• Obtains one of $\frac{f'(x)}{f(x)}$ or $1 + \ln x$, or equivalent merit	1

Sample answer:

$$\log f(x) = x \log x$$

$$\text{Differentiating w.r.t. } x \quad \frac{f'(x)}{f(x)} = 1 + \log x$$

Now $f(x) > 0$, so stationary points occur when $1 + \log x = 0$.

$\therefore x = \frac{1}{e}$. Check gradients each side.

x	$\frac{1}{e}^-$	$\frac{1}{e}$	$\frac{1}{e}^+$
$f'(x)$	–	0	+

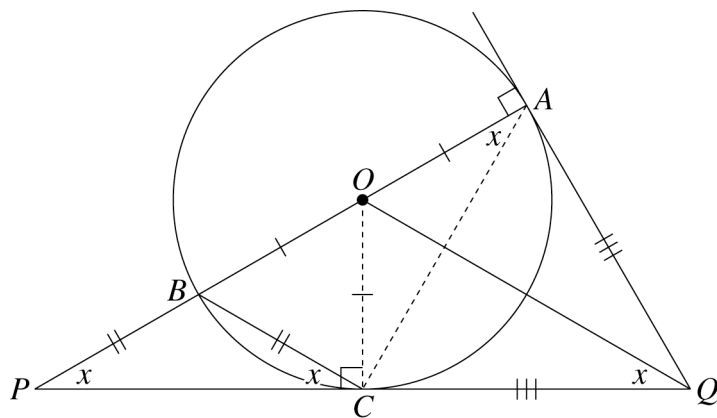
So there is a local minimum at $x = \frac{1}{e}$

Alternatively, $f'(x) = (1 + \log x)f(x)$

$$\text{so } f''(x) = \frac{1}{x}f(x) + (1 + \log x)f'(x)$$

$$\text{and } f''\left(\frac{1}{e}\right) = ef\left(\frac{1}{e}\right) > 0 \text{ so } x = \frac{1}{e} \text{ gives a minimum.}$$

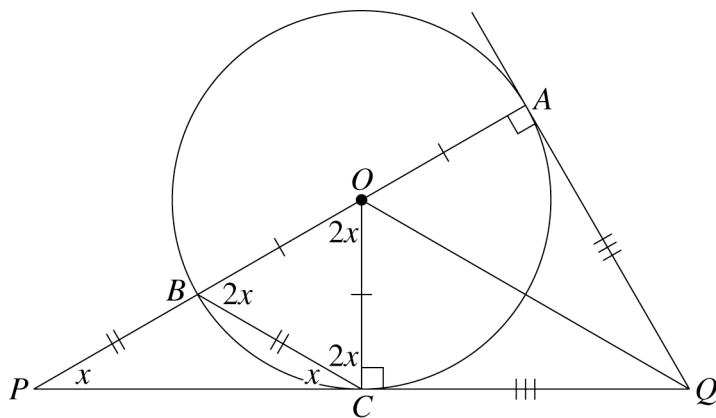
Question 13 (b) (continued)

Alternative solution

$OP = OQ$	Let $\angle BPC = x$	
	$\therefore \angle BCP = x$	(Base $\angle$ s of isosceles $\triangle$)
	Construct AC	
	$\angle BAC = x$	(Alt. seg. thm.)
	Hence, $AQ \perp OA$ and $OC \perp PQ$	(Tangents to circle)
	$OAQC$ is cyclic quad.	(Opp. $\angle$ s supplementary)
	$\therefore \angle OAC = x$	(Angles standing on same arc)
	Hence, $OP = OQ$	(Base $\angle$ s of isosceles $\triangle$ are equal)

Question 13 (b) (continued)

Alternative solution



Let $\angle BPC = x$

$$\angle BPC = \angle PCB = x \quad (\text{Base } \angle \text{ isosceles } \triangle)$$

$$\angle OBC = 2x \quad (\text{Exterior } \angle \triangle PBC)$$

$$\therefore \angle OCB = 2x \quad (\text{Equal } \angle\text{s of isosceles } \triangle OBC, OB = OC \text{ radii})$$

$$\therefore \angle OCP = \frac{\pi}{2} = 3x \quad (\angle \text{ between radius and tangent})$$

$$\therefore x = \frac{\pi}{6}$$

$$\therefore \triangle OBC \text{ is equilateral} \quad (\angle s = 60^\circ)$$

$$\therefore \triangle OCQ \equiv \triangle OAQ \quad (OC = OA \text{ radii, } OQ \text{ common, } AQ = CQ)$$

$$\angle AOC = 4x \quad (\text{Exterior angle})$$

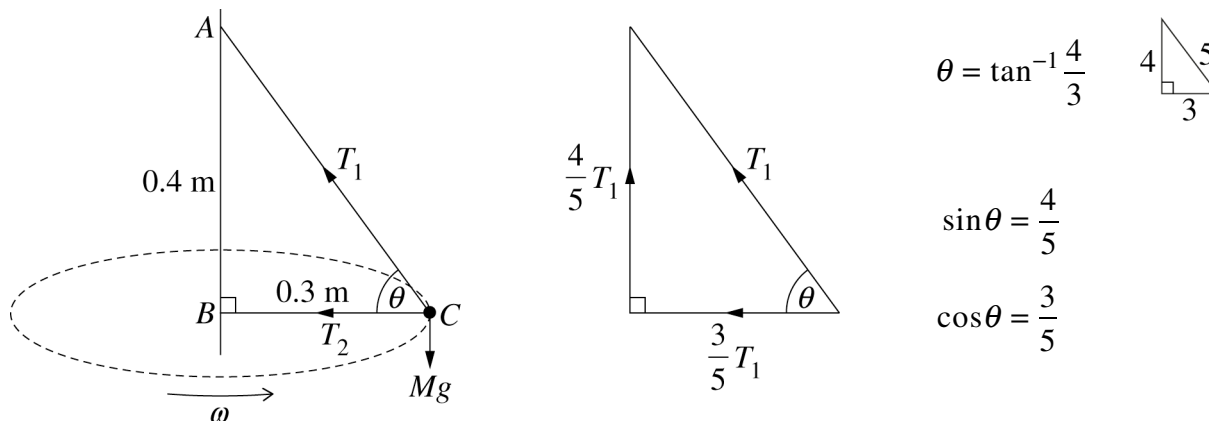
$$\angle COQ = \angle AOQ = 2x \quad (\triangle OCQ \equiv \triangle OAQ)$$

$$\therefore \triangle OPC \equiv \triangle OQC \quad (\text{Equiangular, } OC \text{ common})$$

$$\therefore OP = OQ$$

Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	3
• Attempts to eliminate T_1 , or equivalent merit	2
• Resolves T_1 into components, or equivalent merit	1

Sample answer:

Resolving vertically: $\frac{4}{5}T_1 = Mg$ ($g = 10$)

$$T_1 = \frac{5}{4}M(10)$$

$$\boxed{T_1 = \frac{25M}{2}}$$

Resolving horizontally: $T_2 + \frac{3}{5}T_1 = M(0.3)\omega^2$

$$\therefore T_2 + \frac{3}{5} + \frac{25M}{2} = \frac{3}{10}M\omega^2$$

$$T_2 = \frac{3}{10}M\omega^2 - \frac{3}{10}(25M)$$

$$\boxed{T_2 = \frac{3}{10}M(\omega^2 - 25)}$$

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$T_2 > T_1 \Rightarrow \frac{3}{10}M(\omega^2 - 25) > \frac{25}{2}M$$

$$3\omega^2 - 75 > 125$$

$$3\omega^2 > 200$$

$$\omega^2 > \frac{200}{3}$$

$$|\omega| > \sqrt{\frac{200}{3}} \quad \omega > 0$$

$$\omega > 10\sqrt{\frac{2}{3}}$$

Question 13 (d) (i)

Criteria	Marks
• Provides correct solution	2
• Differentiates $p(x)$, or equivalent merit	1

Sample answer:

$$p(x) = ax^3 + bx^2 + cx + d$$

$$p'(x) = 3ax^2 + 2bx + c$$

$$\Delta = (2b)^2 - 4(3a)(c)$$

$$= 4b^2 - 12ac$$

$$= 4(b^2 - 3ac)$$

$$< 0 \text{ if } b^2 - 3ac < 0 \text{ as given.}$$

$\therefore p'(x)$ has no real solutions.

$\therefore p(x)$ has no stationary points.

Hence $p(x)$ is either an increasing or decreasing function for all x .

$\therefore p(x)$ cuts the x -axis only once.

Question 13 (d) (ii)

Criteria	Marks
• Provides correct solution	2
• Shows $p'\left(-\frac{b}{3a}\right) = 0$, or equivalent merit	1

Sample answer:

$$p\left(-\frac{b}{3a}\right) = 0$$

$$\begin{aligned} p'\left(-\frac{b}{3a}\right) &= 3a\left(-\frac{b}{3a}\right)^2 + 2b\left(-\frac{b}{3a}\right) + c \\ &= \frac{b^2}{3a} - \frac{2b^2}{3a} + c \\ &= -\frac{1}{3a}(b^2 - 3ac) \\ &= 0 \end{aligned}$$

So at least a double root

$$\begin{aligned} p''(x) &= 6ax + 2b \\ p''\left(-\frac{b}{3a}\right) &= 6a\left(-\frac{b}{3a}\right) + 2b \\ &= -2b + 2b \\ &= 0 \end{aligned}$$

So $x = -\frac{b}{3a}$ is a triple root.

Alternatively, $b^2 = 3ac \Rightarrow c = \frac{b^2}{3a}$

$$p(x) = ax^3 + bx^2 + \frac{b^2x}{3a} + d$$

$$p\left(-\frac{b}{3a}\right) = 0$$

$$\Rightarrow -\frac{ab^3}{27a^3} + \frac{b^3}{9a^2} - \frac{b^3}{9a^2} + d = 0$$

$$\Rightarrow d = \frac{b^3}{27a^2}$$

$$\begin{aligned} \therefore 27a^2 p(x) &= 27a^3 x^3 + 27a^2 b x^2 + 9ab^2 x + b^3 \\ &= (3ax + b)^3 \end{aligned}$$

$$\therefore p(x) \text{ has a triple root at } x = -\frac{b}{3a}.$$

Question 14 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

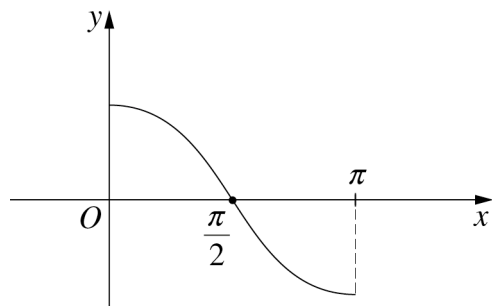
$$\begin{aligned}
 & \int \sin^3 x \, dx \\
 &= \int \sin x (1 - \cos^2 x) \, dx \\
 &= \int \sin x - \sin x \cos^2 x \, dx \\
 &= -\cos x + \frac{\cos^3 x}{3} + C \\
 &= \frac{1}{3} \cos^3 x - \cos x + C
 \end{aligned}$$

$$\begin{aligned}
 \text{OR } & \frac{d}{dx} \left(\frac{1}{3} \cos^3 x - \cos x + C \right) \\
 &= \frac{1}{3} \cdot 3 \cos^2 x (-\sin x) - (-\sin x) + 0 \\
 &= -\cos^2 x \sin x + \sin x \\
 &= \sin x (-\cos^2 x + 1) \\
 &= \sin x (\sin^2 x) \\
 &= \sin^3 x
 \end{aligned}$$

$$\therefore \int \sin^3 x \, dx = \frac{1}{3} \cos^3 x - \cos x + C$$

Question 14 (a) (ii)

Criteria	Marks
• Provides correct explanation	1

Sample answer:

$$\int_0^{\pi} \cos x \, dx = 0 \text{ by symmetry}$$

Since $2n - 1$ is odd,

$$\int_0^{\frac{\pi}{2}} \cos^{2n-1} x \, dx = - \int_{\frac{\pi}{2}}^{\pi} \cos^{2n-1} x \, dx$$

$$\therefore \int_0^{\pi} \cos^{2n-1} x \, dx = 0$$

Question 14 (a) (iii)

Criteria	Marks
• Provides correct solution	3
• Attempts to evaluate correct integral using integration by parts, or equivalent merit	2
• Obtains correct expression for the volume, or equivalent merit	1

Sample answer:

By method of shells,

$$\begin{aligned}\text{Volume} &= 2\pi \int x f(x) dx \\ &= 2\pi \int_0^{\pi} x \sin^3 x dx\end{aligned}$$

Using integration by parts and part (i)

$$\text{Volume} = 2\pi \left(\left[x \left(\frac{1}{3} \cos^3 x - \cos x \right) \right]_0^{\pi} - \int_0^{\pi} \left(\frac{1}{3} \cos^3 x - \cos x \right) dx \right)$$

The integral above is 0 by part (ii), so

$$\begin{aligned}\text{Volume} &= 2\pi \left(\pi \left(-\frac{1}{3} + 1 - 0 \right) - 0 \right) \\ &= \frac{4}{3} \pi^2\end{aligned}$$

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	3
• Obtains an integral that can be easily evaluated, or equivalent merit	2
• Chooses suitable substitution, or equivalent merit	1

Sample answer:

$$I_0 = \int_0^1 \frac{1}{(x^2 + 1)^2} dx$$

Let $x = \tan \theta$, then $\frac{dx}{d\theta} = \sec^2 \theta$, so

$$I_0 = \int_0^{\frac{\pi}{4}} \frac{\sec^2 \theta}{\sec^4 \theta} d\theta$$

$$= \int_0^{\frac{\pi}{4}} \cos^2 \theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{1 + \cos 2\theta}{2} d\theta$$

$$= \frac{1}{2} \theta + \frac{\sin 2\theta}{4} \Big|_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{8} + \frac{1}{4}$$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\begin{aligned} I_0 + I_2 &= \int_0^1 \frac{1+x^2}{(x^2+1)^2} dx \\ &= \int_0^1 \frac{1}{x^2+1} dx \\ &= \tan^{-1}x \Big|_0^1 = \frac{\pi}{4} \end{aligned}$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	3
• Obtains the value of $I_2 + I_4$, or equivalent merit	2
• Attempts to evaluate $I_2 + I_4$, or equivalent merit	1

Sample answer:Consider $I_2 + I_4$

$$= \int_0^1 \frac{x^2 + x^4}{(x^2 + 1)^2} dx$$

$$= \int_0^1 \frac{x^2}{x^2 + 1} dx$$

$$= \int_0^1 1 - \frac{1}{x^2 + 1} dx$$

$$= x - \tan^{-1} x \Big|_0^1$$

$$= 1 - \frac{\pi}{4}$$

$$\therefore I_4 = 1 - \frac{\pi}{4} - I_2$$

$$= 1 - \frac{\pi}{4} - \left(\frac{\pi}{4} - I_0 \right)$$

$$= 1 - \frac{\pi}{2} + \left(\frac{\pi}{8} + \frac{1}{4} \right)$$

$$= \frac{5}{4} - \frac{3\pi}{8}$$

Question 14 (c)

Criteria	Marks
• Provides correct solution	3
• Observes $(x-1)(\sqrt{x}-1) \geq 0$ for $x \geq 1$, or equivalent merit	2
• Investigates $x\sqrt{x}+1-x-\sqrt{x}$, or equivalent merit	1

Sample answer:

For $x \geq 0$, Required to show that $x\sqrt{x}+1 \geq x+\sqrt{x}$

$$\begin{aligned}
 \text{Now } x\sqrt{x}+1-(x+\sqrt{x}) &= x\sqrt{x}-\sqrt{x}-x+1 \\
 &= \sqrt{x}(x-1)-(x-1) \\
 &= (x-1)(\sqrt{x}-1)
 \end{aligned}$$

$$\begin{aligned}
 \text{For } 0 \leq x < 1, \quad (x-1) < 0, \quad (\sqrt{x}-1) < 0 \quad & (-)(-) \\
 \therefore x\sqrt{x}+1-(x+\sqrt{x}) > 0
 \end{aligned}$$

$$\begin{aligned}
 \text{For } x \geq 1, \quad (x-1) \geq 0, \quad (\sqrt{x}-1) \geq 0 \quad & (+)(+) \\
 \therefore x\sqrt{x}+1-(x+\sqrt{x}) \geq 0
 \end{aligned}$$

Hence for $x \geq 0$, $x\sqrt{x}+1 \geq x+\sqrt{x}$

Alternative solution

For $x \geq 0$, Required to show that $x\sqrt{x}+1 \geq x+\sqrt{x}$

$$\begin{aligned}
 \text{Now } x\sqrt{x}+1-(x+\sqrt{x}) &= x\sqrt{x}-\sqrt{x}-x+1 \\
 &= \sqrt{x}(x-1)-(x-1) \\
 &= (x-1)(\sqrt{x}-1) \\
 &= (\sqrt{x}+1)(\sqrt{x}-1)(\sqrt{x}-1) \\
 &= (\sqrt{x}+1)(\sqrt{x}-1)^2 \\
 &\geq 0
 \end{aligned}$$

Since $\sqrt{x}+1 \geq 0$ for $x \geq 0$

and $(\sqrt{x}-1)^2 \geq 0$ for real $x \geq 0$

$$\therefore x\sqrt{x}+1 \geq x+\sqrt{x}$$

Question 15 (a)

Criteria	Marks
• Provides correct solution	2
• Obtains the value of $\alpha^2 + \beta^2 + \gamma^2$, or equivalent merit	1

Sample answer:

$x^3 - 3x + 1 = 0$ has roots α, β, γ , then

$$\alpha + \beta + \gamma = 0$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = -3$$

$$\alpha\beta\gamma = -1$$

Let $A = \alpha^2, B = \beta^2, C = \gamma^2$

$$A + B + C = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) = 6$$

$$\begin{aligned} AB + AC + BC &= \alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 \\ &= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma) \\ &= 9 \end{aligned}$$

$$ABC = \alpha^2\beta^2\gamma^2 = 1$$

$\therefore A, B, C$ satisfy $x^3 - 6x^2 + 9x - 1$

Alternative solution

Let $y = x^2$ then, $x = \sqrt{y}$

So $(\sqrt{y})^3 - 3\sqrt{y} + 1 = 0$

$$\Rightarrow y\sqrt{y} - 3\sqrt{y} = -1$$

$$\sqrt{y}(y - 3) = -1$$

$$y(y - 3)^2 = 1$$

$$\Rightarrow y^3 - 6y^2 + 9y - 1 = 0$$

Question 15 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Correctly applies $a = \frac{d\left(\frac{1}{2}v^2\right)}{dx}$, or equivalent merit	1

Sample answer:

$$\ddot{x} = -\frac{\mu^2}{x^2}$$

$$\frac{1}{2} \frac{d}{dx}(v^2) = -\frac{\mu^2}{x^2}$$

$$\Rightarrow v^2 = \frac{2\mu^2}{x} + C$$

$$x = b, v = 0 \Rightarrow C = -\frac{2\mu^2}{b}$$

$$\therefore v^2 = 2\mu^2 \left(\frac{1}{x} - \frac{1}{b} \right)$$

$$= 2\mu^2 \left(\frac{b-x}{bx} \right)$$

$$\therefore v = -\mu\sqrt{2} \sqrt{\frac{b-x}{bx}}, \text{ negative root since the particle moves to the left.}$$

Question 15 (b) (ii)

Criteria	Marks
• Provides correct solution	3
• Correctly applies given substitution to integrand, or equivalent merit	2
• Obtains integral for t in terms of x , or equivalent merit	1

Sample answer:

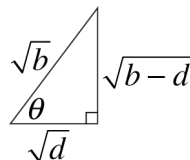
$$\frac{dx}{dt} = -\mu\sqrt{2} \sqrt{\frac{b-x}{bx}}$$

$$t = -\frac{1}{\mu\sqrt{2}} \int_b^d \sqrt{\frac{bx}{b-x}} dx$$

Put $x = b \cos^2 \theta$, $\frac{dx}{d\theta} = -2b \cos \theta \sin \theta$

$$x = b \Rightarrow \theta = 0$$

$$x = d \Rightarrow \theta = \cos^{-1} \sqrt{\frac{d}{b}}$$



$$\text{Then } t = \frac{2b\sqrt{b}}{\sqrt{2}\mu} \int_0^{\cos^{-1} \sqrt{\frac{d}{b}}} \frac{\cos \theta}{\sin \theta} \cdot \cos \theta \sin \theta d\theta$$

$$= \frac{b\sqrt{2b}}{\mu} \int_0^{\cos^{-1} \sqrt{\frac{d}{b}}} \cos^2 \theta d\theta$$

Question 15 (b) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\begin{aligned}\lim_{d \rightarrow 0^+} t &= \frac{1}{\mu} \sqrt{\frac{b}{2}} \cdot \frac{b\pi}{2} \\ &= \frac{\pi}{2\sqrt{2}\mu} b\sqrt{b}\end{aligned}$$

Question 15 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Uses method of partial fractions to compare numerators, or equivalent merit	1

Sample answer:

Write $\frac{3!}{x(x+1)(x+2)(x+3)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2} + \frac{D}{x+3}$

$$\therefore 3! = A(x+1)(x+2)(x+3) + Bx(x+2)(x+3) + Cx(x+1)(x+3) + Dx(x+1)(x+2)$$

$$x = 0 \Rightarrow A = 1$$

$$x = -1 \Rightarrow -2B = 6 \Rightarrow B = -3$$

$$x = -2 \Rightarrow 2C = 6 \Rightarrow C = 3$$

$$x = -3 \Rightarrow -6D = 6 \Rightarrow D = -1$$

Question 15 (c) (ii)

Criteria	Marks
• Provides correct solution	3
• Uses $x = -k$ to obtain an expression for a_k , or equivalent merit	2
• Uses method of partial fractions to compare numerators, or equivalent merit	1

Sample answer:

$$\frac{n!}{x(x+1)\dots(x+n)} = \frac{a_0}{x} + \frac{a_1}{x+1} + \dots + \frac{a_k}{x+k} + \dots + \frac{a_n}{x+n}$$

Put RHS over common denominator, then general term is

$$a_k x(x+1)\dots(x+k-1)(x+k+1)\dots(x+n)$$

At $x = -k$, all terms are zero except,

$$-k(-k+1)\dots(-1)(1)2\dots(-k+n)a_k \equiv n!$$

$$(-1)^k [1 \cdot 2 \dots k][1 \cdot 2 \dots (n-k)]a_k = n!$$

$$\therefore a_k = (-1)^k \frac{n!}{k!(n-k)!}$$

$$= (-1)^k \binom{n}{k}$$

Question 15 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Identifies $x = 1$ using part (ii), or equivalent merit	1

Sample answer:Put $x = 1$ in part (ii) then

$$\begin{aligned} \frac{1}{1} - \frac{1}{2} \binom{n}{1} + \frac{1}{3} \binom{n}{2} - \frac{1}{4} \binom{n}{3} + \dots + \frac{(-1)^n}{n+1} &= \frac{n!}{1 \cdot 2 \dots (n+1)} \\ &= \frac{n!}{(n+1)!} = \frac{1}{n+1} \end{aligned}$$

Hence the limiting sum $= \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$.

Question 16 (a) (i)

Criteria	Marks
• Provides correct solution	3
• Finds θ or α , or equivalent merit	2
• Uses real and imaginary parts or equivalent merit	1

Sample answer:

$$z = \cos \theta + i \sin \theta$$

$$w = \cos \alpha + i \sin \alpha$$

$$1 + z + w = 0$$

$$\Rightarrow (1 + \cos \theta + \cos \alpha) + i(\sin \theta + \sin \alpha) = 0$$

Equating real and imaginary parts,

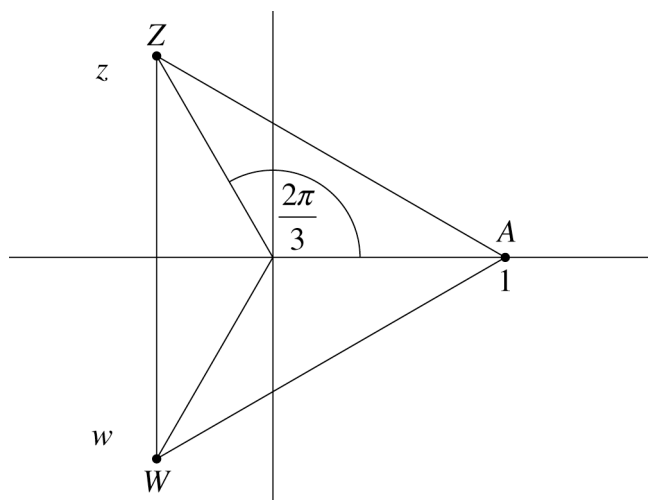
$$1 + \cos \theta + \cos \alpha = 0 \quad (1)$$

$$\sin \theta = -\sin \alpha \quad (2)$$

$$(2) \Rightarrow \theta = -\alpha$$

$$(1) \Rightarrow \cos \theta = -\frac{1}{2}$$

$$\therefore \theta = \frac{2\pi}{3}, \quad \alpha = -\frac{2\pi}{3}$$



Since $|z| = |w| = 1$

$$\angle ZAW = \frac{\pi}{3}$$

So the triangle is equilateral.

Question 16 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Divides by one of $2i$, z_1 , z_2 , or equivalent merit	1

Sample answer:

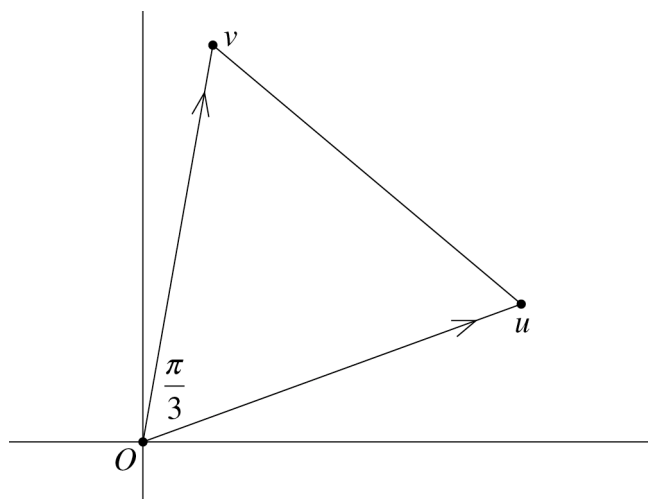
Dividing each complex number by $2i$, we have 1 , $\frac{z_1}{2i}$, $\frac{z_2}{2i}$, which have unit modulus and sum 0.

So by part (i) they form the vertices of an equilateral triangle.

Multiplying by $2i$ rotates the triangle by $\frac{\pi}{2}$ and dilates the sides by a factor of 2, giving a similar triangle. Hence $2i$, z_1 , z_2 form the vertices of an equilateral triangle.

Question 16 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Observes that v is u rotated by $\pm\frac{\pi}{3}$	1

Sample answer:

The vector O_v is vector O_u rotated about O anticlockwise through $\frac{\pi}{3}$, so

$$v = \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) u$$

$$\therefore v^3 = -u^3$$

$$\Rightarrow (v+u)(u^2 + v^2 - uv) = 0$$

$$v \neq -u, \text{ so } u^2 + v^2 = uv$$

Question 16 (b) (ii)

Criteria	Marks
• Provides a correct example	1

Sample answer:

For example, take $u = 1$ then

$$1 + v^2 = v, \text{ so}$$

$$v^2 - v + 1 = 0$$

$$v = \frac{1 + \sqrt{-3}}{2} = \frac{1 + \sqrt{3}i}{2}$$

Alternative Take $u = 1$
then can use

$$\begin{aligned} v &= \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \\ &= \frac{1}{2} + \frac{i\sqrt{3}}{2} \end{aligned}$$

Question 16 (c) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

There are $(n - 1)$ people who can swap hats with Tom. Removing Tom and this person then $(n - 2)$ people are left, giving $D(n - 2)$ derangements.

Hence there are $(n - 1)D(n - 2)$ such derangements.

Question 16 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Identifies remaining case, or equivalent merit	1

Sample answer:

There are $(n-1)$ people who can take Tom's hat. Suppose Tom has not directly swapped a hat with exactly one person. One person takes Tom's hat and leaves. Tom takes their place and we have a derangement of $(n-1)$ people. Hence in this case there are $(n-1)D(n-1)$ derangements.

Hence the total number of derangements is $D(n) = (n-1)[D(n-1) + D(n-2)]$.

Alternative

Ignoring Tom, there are $D(n-1)$ derangements of the remaining hats. If Tom swaps hats with any of the $(n-1)$ others we have a suitable derangement. A total of $(n-1)D(n-1)$ derangements.

Tom either swaps hats with one _____ $(n-1)D(n-2)$

or he doesn't _____ $(n-1)D(n-1)$

Total $D(n) = (n-1)[D(n-1) + D(n-2)]$

Question 16 (c) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

From part (ii),

$$D(n) = (n-1)[D(n-1) + D(n-2)]$$

$$\therefore D(n) = nD(n-1) - D(n-1) + (n-1)D(n-2)$$

$$\therefore D(n) - nD(n-1) = -[D(n-1) - (n-1)D(n-2)]$$

Question 16 (c) (iv)

Criteria	Marks
• Provides correct solution	1

Sample answer:

By part (iii),

$$\begin{aligned}
 D(n) - nD(n-1) &= -[D(n-1) - (n-1)D(n-2)] \\
 &= (-1)^2 [D(n-2) - (n-2)D(n-3)] \\
 &= (-1)^3 [D(n-3) - (n-3)D(n-4)] \\
 &\quad \vdots \\
 &= (-1)^{n-2} [D(2) - D(1)] \\
 &= (-1)^{n-2} [1 - 0] \\
 &= (-1)^{n-2} \times (-1)^2 \\
 &= (-1)^n
 \end{aligned}$$

Question 16 (c) (v)

Criteria	Marks
• Provides correct solution	2
• Makes some progress	1

Sample answer:When $n = 1$

$$1! \sum_{r=0}^1 \frac{(-1)^r}{r!} = 1!(1-1) = 0 = D(1)$$

So statement is true for $n = 1$.Let k be an integer such that $D(k) = k! \sum_{r=0}^k \frac{(-1)^r}{r!}$

Then

$$D(k+1) = (k+1)D(k) + (-1)^{k+1} \quad (\text{By part (iv)})$$

$$= (k+1)! \sum_{r=0}^k \frac{(-1)^r}{r!} + (-1)^{k+1} \cdot \frac{(k+1)!}{(k+1)!}$$

$$= (k+1)! \sum_{r=0}^{k+1} \frac{(-1)^r}{r!}$$

 $\therefore$ Statement is true for $n = k+1$, so true for all $n \geq 1$ by induction.Alternative solution

From part (iv),

$$\frac{1}{n!}D(n) - \frac{1}{(n-1)!}D(n-1) = \frac{(-1)^n}{n!}$$

Substituting $n = 2, 3, \dots$ and adding we have

$$\frac{1}{n!}D(n) - D(1) = \sum_{r=2}^n \frac{(-1)^r}{r!}$$

Now since $\frac{(-1)^0}{0!} + \frac{(-1)^1}{1!} = 0$ we have

$$D(n) = n! \sum_{r=0}^n \frac{(-1)^r}{r!}$$

2016 HSC Mathematics Extension 2

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	1.3	E6
2	1	7.2	E4
3	1	3.4	E3
4	1	2.1, 2.2, 2.3	E3
5	1	2.1, 2.2	E3
6	1	7.4, 8 (16.3E)	PE3, E4
7	1	3.3	E4
8	1	6.3	E5
9	1	5.1	E7
10	1	2.4	E3

Section II

Question	Marks	Content	Syllabus outcomes
11 (a) (i)	2	2.2	E3
11 (a) (ii)	1	2.4	E3
11 (a) (iii)	1	2.4	E3
11 (b)	3	4.1	E8
11 (c)	2	1.8	E6
11 (d) (i)	2	1.7	E6
11 (d) (ii)	2	1.5	E6
11 (e)	2	8 (15E)	HE4, E6
12 (a) (i)	1	3.1	E3
12 (a) (ii)	1	3.1	E3
12 (a) (iii)	1	3.1	E3
12 (a) (iv)	1	3.1	E3
12 (b) (i)	1	8 (8.8)	PE5, E2
12 (b) (ii)	2	4.1	E8
12 (c) (i)	2	2.1, 2.4, 8 (17.1E)	E3
12 (c) (ii)	1	8	E3
12 (d) (i)	2	3.3	E4
12 (d) (ii)	3	3.3	E4
13 (a)	3	1.8, 8 (12.4)	E6
13 (b)	4	8 (2.10E)	E2, PE3

Question	Marks	Content	Syllabus outcomes
13 (c) (i)	3	6.3.3	E5
13 (c) (ii)	1	6.3.3	E5
13 (d) (i)	2	7.5	E4
13 (d) (ii)	2	7.2, 7.5	E4
14 (a) (i)	1	4.1	E8
14 (a) (ii)	1	1.6, 1.9	E6
14 (a) (iii)	3	5.1	E7
14 (b) (i)	3	4.1	E8
14 (b) (ii)	1	4.1	E8
14 (b) (iii)	3	4.1	E8
14 (c)	3	8.3	E4
15 (a)	2	7.5	E4
15 (b) (i)	2	6.1	E5
15 (b) (ii)	3	4.1	E8
15 (b) (iii)	1	6.1	E2
15 (c) (i)	2	7.6	E8
15 (c) (ii)	3	7.6	E8
15 (c) (iii)	2	8 (17.3E)	E4
16 (a) (i)	3	2.1, 2.2	E3
16 (a) (ii)	2	2.1, 2.2	E3
16 (b) (i)	2	2.1, 2.2	E3
16 (b) (ii)	1	2.1, 2.2	E3
16 (c) (i)	1	8 (18E)	E2
16 (c) (ii)	2	8 (18E)	E2
16 (c) (iii)	1	8 (18E)	E2
16 (c) (iv)	1	8 (18E)	E2
16 (c) (v)	2	8 (18E)	E2